Surface Tension
&
Energy

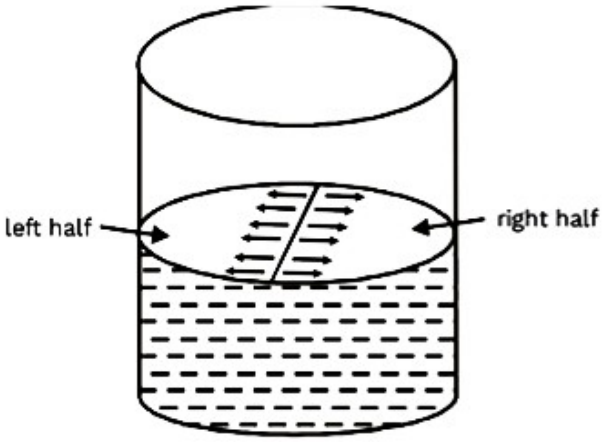
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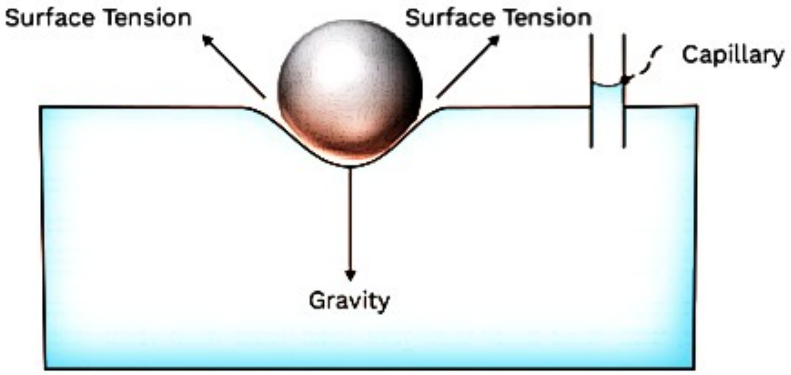
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Surface Tension & Energy

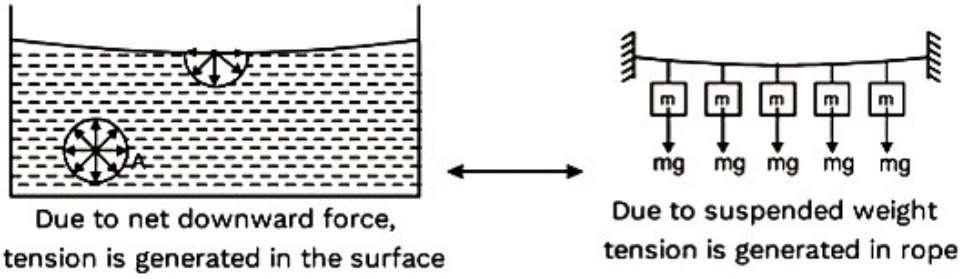
Surface Tension is a property by which liquid surface behaves like a stretched membrane (Rubber sheet) and tries to minimize it's area.

Force due to surface Tension: When a liquid or two immiscible liquids are in contact, an unbalanced force is developed at the interface stretched over the entire fluid mass. The intensity of molecular attraction per unit length along any line in the surface is called force due to surface Tension.





Explanation of surface tension based on intermolecular forces:



Due to net downward force,
a tension is generated in the surface

Due to suspended weight
tension is generated in rope

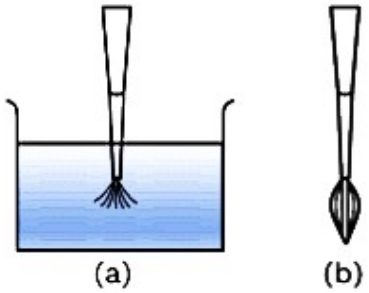
Cohesive forces (attractive forces between molecules of same substances) cause surface tension in liquids.
Consider a container filled with a liquid and two molecules, one inside the liquid and other on the free surface of the liquid.
The inner liquid molecule experiences equal cohesive forces from all directions and net cohesive force on this molecule will be zero.
But, for the molecule on the free surface of the liquid, only the liquid molecules below it are applying cohesive forces while its upper half is free. So, resultant cohesive force on a molecule on the free surface of a liquid is downwards.
So, for each molecule on the free surface, there is a net downward cohesive force. This net downward cohesive force generates a tension in the free surface just like suspended weights from a rope create tension in the rope.

Explanation of surface tension based on energy:

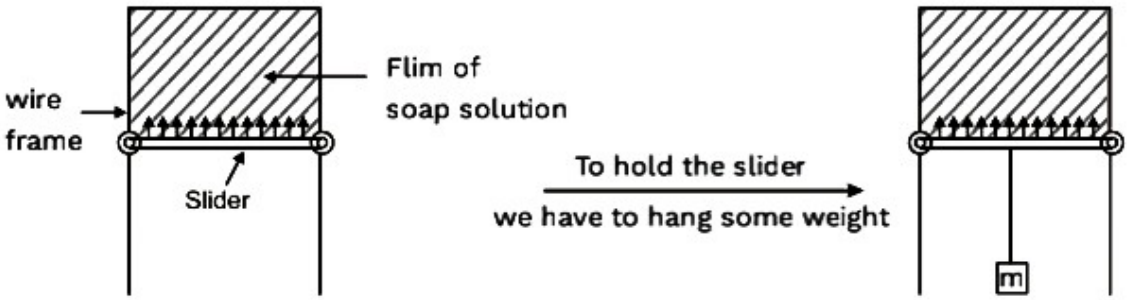
A liquid molecule inside the liquid is surrounded by more liquid molecules as compared to another liquid molecule on the free surface. So, the inner molecule has more negative energy as compared to the free surface molecule.
Less negative means more energy which leads to say that a free surface molecule has more energy than a inner liquid molecule.
To attain stability, energy should be minimum possible. To attain minimum possible energy, the free surface area should be minimum. Thus the free surface tries to minimize its surface area in order to attain stability and it causes a tension in the free surface.

Some Observed Phenomena Based on surface Tension:

1. Insects, like mosquitoes, move about freely on the surface of ponds without sinking. This is on account of the support they get from the stretched surface film of water.
2. If we gently place a greased needle on the free surface of Water at rest, we shall find that the needle does not sink. A careful observation shows that the surface of water below the needle gets slightly depressed. This shows that the free surface of the liquid layer supports the needle in the same Way as a stretched rubber membrane does.
3. Hair Of a brush, when dipped in water spread out Fig. (a); but stick together on taking the brush out as shown in Fig. (b).



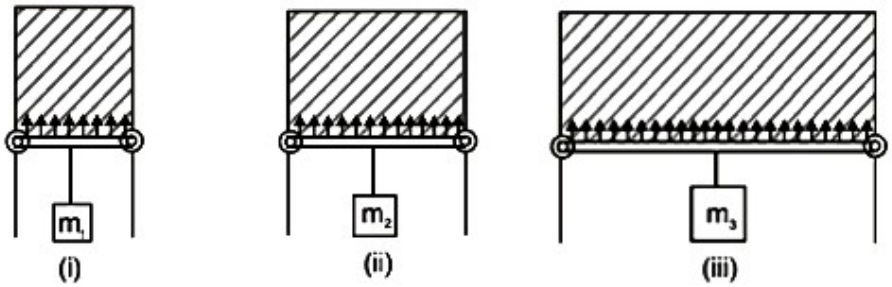
Determination of force due to surface tension:



In above figure, a fixed wire frame having a U shape is taken. A very light slider is fitted on this frame. This frame is put inside a soap solution and taken out. There is a thin film of soap solution formed between the slider and frame.

It has been observed that on releasing the slider, it moves upwards. It shows that the soap solution film has tension.

On hanging some weight to the slider, the upward surfce tension can be balanced and in equilibrium, we can measure the surface tension of the soap solution.



It has been found experimentally that,

Surface tension force $(F) \propto$ contact length (ℓ)

$$F = (T)\,\ell$$



Here T is the surface tension.

T depends on-

a. Properties of the liquid, and

b. On the medium around the liquid.

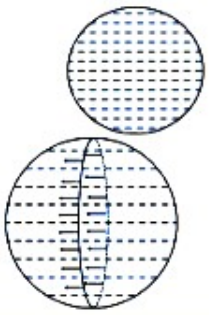
- Surface tension (T) decreases on increasing temperature.
- Surface tension (T) increases on adding highly soluble substances (like NaCl).
- Surface tension (T) decreases on adding sparingly soluble substance (like soap).

Force due to surface tension, F = T l

Then, $T = \frac{F}{\ell}$

Thus, surface tension can also be defined as the force per unit length on a fictitious line on the free surface of the liquid acting perpendicular to the line and in the same plane as that of the free surface of the liquid.

Q. **Find the force due to surface tension acting between left and right half surfaces of a water drop of radius R if the surface tension of liquid is T.**



Sol. Length of the boundary in contact between the left and right half will be equal to the perimeter of the liquid drop.

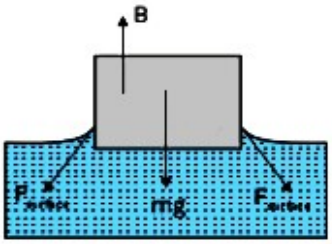
So, F = (T)(2πR)

Q. **A cube of mass m = 800g floats on the surface of water. Water wets it completely. The cube is 10 cm on each edge. By what additional distance is it buoyed up or down by surface tension? Surface tension of water = 0.07 Nm^{-1}.**

Sol. If surface tension is neglected the condition for floating gives

$800 \times 10^{-3} g = (0.1^2 \times \rho) g$

or $x = 0.08$ m ($\rho = 1000\ kgm^{-3}$ for water)

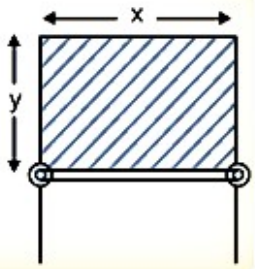


Since water wets the cube, the angle of contact is zero and force of surface tension acts vertically downwards. So, it is buoyed down by surface tension.

$$\therefore \quad 800 \times 10^{-3} g + 4 \times 0.1 \times 0.07 = \left(0.1^2 x' \rho\right) g$$

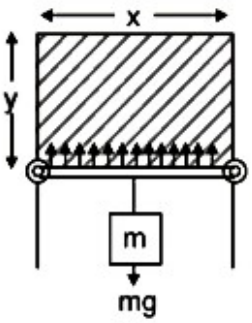
$$\text{or} \quad x' = 0.08 + \frac{0.028}{98} = 0.08 + 2.8 \times 10^{-4}\ \text{m}$$

Q. **A thin soap solution filem havinng length x and width y is formed between a thin U shaped wire frame and a slider. What will be the force due to surface tension on the slider if the surface tension of the soap solution is T.**
Also, find the wieght needed to be suspended from the slider to keep it in equilibrium.



Sol. The soap solution surface formed within the frame will pull the slider upwards with a force,

$$F = (T \times 2x)$$



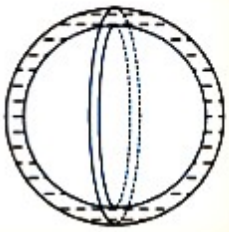
The contact length has been taken as 2x here because a thin film has two free surfaces.

At equilibrium of the slider, the surface tension force is balanced by the weight of the suspended block.

So, (T) (2x) = mg

Mass of the block suspended, $m = \frac{2Tx}{g}$

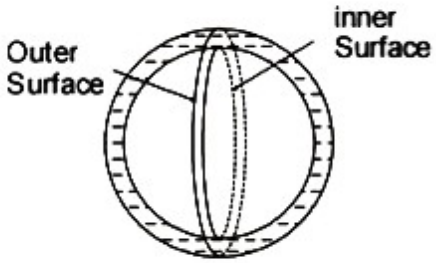
Q. **Find the force due to surface tension between the left and right half of a soap bubble if the surface tension of the soap solution is T?**



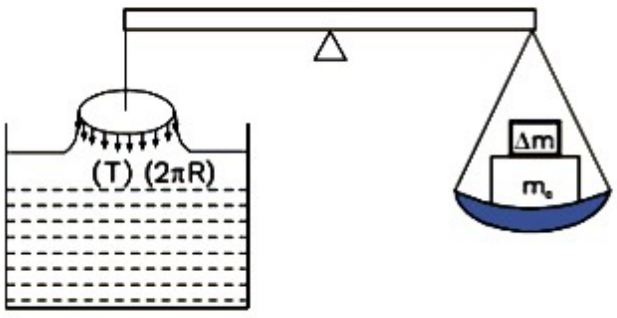
Sol. The soap bubble has two free surfaces.

So, net force due to surface tension, $F = (T)(2\pi R) + (T)(2\pi R)$

$= T(4\pi R)$



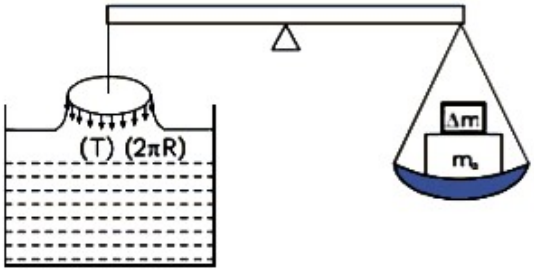
Q. **A balance has its one arm comprised of a thin disc whose surface is touching a liquid as shown in the figure. A weight on the other arm of the balance is balancing the disc of radius R. What is the extra weight added to the other side of the balance to take the disc out of water?**



Sol. Force due to surface tension on the disc = $(T)(2\pi R)$

For disc to just come out, $(T)(2\pi R) = (\Delta m)g \Rightarrow \Delta m = \dfrac{(T)(2\pi R)}{g}$

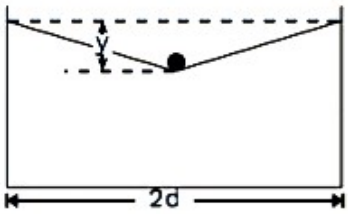
Q. **If in the previous case, a ring of inner radius R_1 and outer radius R_2 is used, then what is the extra weight added to just pull out the ring out of water?**



Sol. Force due to surface tension on the ring = $(T)2\pi(R_1 + R_2)$

For ring to just come out, $(T)2\pi(R_1 + R_2) = (\Delta m)g \Rightarrow \Delta m = \dfrac{(T)2\pi(R_1 + R_2)}{g}$

Q. **In a container of width 2d containing a liquid, a straight uniform wire of negligible radius is kept on the free surface. As a result, the surface of the liquid gets depressed by a vertical distance y (y<<d) at the centre as shown. If λ is the mass per unit length of the wire, then find the surface tension of the liquid kept in the container.**



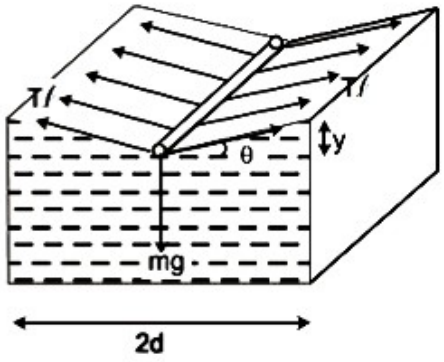
Sol. Balancing upward and downward forces

$$2\left(T\ell \sin\theta\right) = \left(\lambda\ell\right)g$$

as angle is very small $\sin\theta \approx \tan\theta = \dfrac{y}{d}$

$$2\left(T\ell\right)\frac{y}{d} = \left(\lambda\ell\right)g$$

$$T = \frac{\lambda g d}{2y}$$

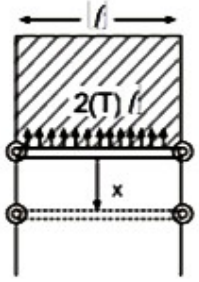


SURFACE ENERGY:

Surface energy is the potential energy stored due to the force of surface tension.

For a thin soap film formed in a U-shaped wire and a slider, net force on the slider is 2Tl.

On moving the slider in forward direction by a distance x, work done by force due to surface tension = - (2Tl) (x)



⇒ Work done against the force due to surface tension = +(2Tl) (x)

⇒ Increase in surface potential energy = +(2T ℓ)x

So, the increase in the potential energy of the surface = +(2Tl) (x)

= +(T) (2l x)

where (2l x) = increase in the surface area

(increase in front area = (l x), increase in back area = (l x))

⇒ Increase in surface potential energy $\Delta U = (T)(\Delta A) = (T)$ (increase in surface area)

or generally, we can say that

$$\boxed{\text{Surface energy } U = (T)(A) = (T)(\text{surface area})}$$

Also $T = \frac{U}{A}$ and previously we have seen that $T = \frac{F}{\ell}$

So, *Surface tension is surface energy per unit surface area Surface tension is also tension force generated on the surface per unit length.*

Q. **What is the decrease in the surface energy when 1000 small water drop each having a radius r combine to form a bigger drop.**

Sol. Suppose radius of big drop is R. During this process, mass will be conserved, so volume will also be conserved.

$$(\text{Volume})_{\text{initial}} = (\text{Volume})_{\text{final}}$$

$$\left(\frac{4}{3}\pi r^3\right) \times 1000 = \left(\frac{4}{3}\pi R^3\right) \Rightarrow R = 10r$$

Loss in surface energy

$$\Delta U_{\text{loss}} = T\Delta A_{\text{loss}} = T\left(\left(4\pi r^2\right) \times 1000 - 4\pi\left(10r\right)^2\right)$$

$$\Delta U_{\text{loss}} = (T)\left(900 \times 4\pi r^2\right)$$

Q. **If a number of little droplets of water, each of radius r, coalesce to form a single drop of radius R, show that the rise in temperature will be given by.**

$$\frac{3T}{J}\left(\frac{1}{r}-\frac{1}{R}\right)$$

where T is the surface tension of water and J is the mechanical equivalent of heat. Here r, R and T are in CGS system.

Sol. Suppose n small water drop combine and form a big drop. During this process volume will be conserved

$$(\text{Volume})_{\text{initial}} = (\text{Volume})_{\text{final}}$$

$$\left(\frac{4}{3}\pi r^3\right)\times n = \frac{4}{3}\pi R^3 \Rightarrow n = \frac{R^3}{r^3}$$

Loss in surface energy

$$\Delta U_{\text{loss}} = T\Delta A_{\text{loss}} = T\left(4\pi r^2 \times n - 4\pi R^2\right)$$

$$\text{Put } n = \frac{R^3}{r^3}$$

$$\text{we get } \Delta U_{\text{loss}} = T\left(4\pi r^2 \times \frac{R^3}{r^3} - 4\pi R^2\right)$$

$$\Rightarrow \quad \Delta U_{\text{loss}} = T4\pi R^3\left(\frac{1}{r}-\frac{1}{R}\right)$$

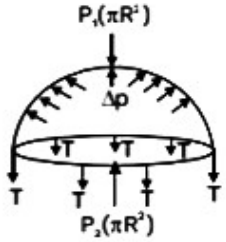
$$\Rightarrow \quad ms\Delta\theta = T\left(4\pi R^3\right)\left(\frac{1}{r}-\frac{1}{R}\right) \qquad \text{where } m = \rho(\text{vol})$$

$$\Rightarrow \quad \left(\frac{1gm}{cm^3}\right)\left(\frac{4}{3}\pi R^3\right)\left(\frac{1cal}{gm^\circ C}\right)\Delta\theta = T\left(4\pi R^3\right)\left(\frac{1}{r}-\frac{1}{R}\right)$$

$$\Rightarrow \quad \Delta\theta = \frac{3T}{J}\left[\frac{1}{r}-\frac{1}{R}\right]$$

Excess Pressure inside A liquid Drop:

Figure of one-half cross section of an air bubble formed inside liquid is drawn.



It is in equilibrium under the action of three forces:

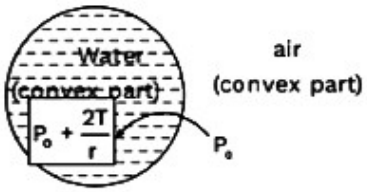
1. due to external pressure P_1
2. due to internal pressure P_2
3. due to surface tension of the liquid T

If R is radius of the bubble, then the force due to external and internal pressures are $P_1(\pi R^2)$ and $P_2(\pi R^2)$ respectively. Since, the surface tension acts around the circumference of bubble, therefore, the force of surface tension is $T(2\pi R)$.

Thus, from the condition of equilibrium,

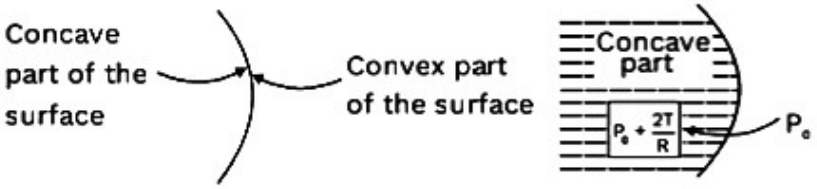
$$P_2\left(\pi R^2\right) = P_1\left(\pi R^2\right) + T\left(2\pi R\right)$$

$$\Rightarrow \boxed{P_2 - P_1 = \frac{2T}{R}}$$



So, pressure inside the drop will be greater than pressure outside the drop by $\frac{2T}{r}$

Generally, we can say that pressure at concave part will be greater than pressure at convex part by $\frac{2T}{r}$ where r is radius of curvature of the surface between them.



Q. **A minute spherical air bubble is rising slowly through a column of mercury contained in a deep jar. If the radius of the bubble at a depth of 100 cm is 0.1 mm, calculate its depth where its radius is 0.126 mm, given that the surface tension of mercury is 567 dyne/cm. Assume that the atmospheric pressure is 76 cm of mercury.**

Sol. The total pressure inside the bubble at depth h_1 is (P is atmospheric pressure)

$$= \left(P + h_1\rho g\right) + \frac{2T}{r_1} = P_1$$

And the total pressure inside the bubble at depth h_2 is $= \left(P + h_2\rho g\right) + \dfrac{2T}{r_2} = P_2$

Now, according to Boyle's Law

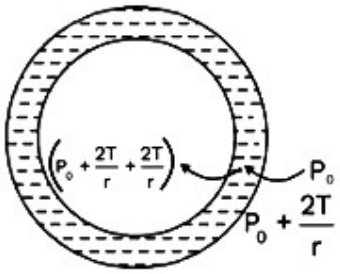
$$P_1V_1 = P_2V_2 \text{ where } V_1 = \frac{4}{3}\pi r_1^3, \text{ and } V_2 = \frac{4}{3}\pi r_2^3$$

Hence, we get $\left[\left(P + h_1\rho g\right) + \dfrac{2T}{r_1}\right]\dfrac{4}{3}\pi r_1^3 = \left[\left(P + h_2\rho g\right) + \dfrac{2T}{r_2}\right]\dfrac{4}{3}\pi r_2^3$

$$\Rightarrow \quad \left[\left(P + h_1\rho g\right) + \frac{2T}{r_1}\right]r_1^3 = \left[\left(P + h_2\rho g\right) + \frac{2T}{r_2}\right]r_2^3$$

Given that: $h_1 = 100$ cm, $r_1 = 0.1$ mm, $r_2 = 0.126$ mm $= 0.126$ cm, $T = 567$ dyne/cm, $P = 76$ cm of mercury, substituting all the values we get $h_2 = 9.48$ cm.

Excess Pressure inside a liquid bubble kept in air:

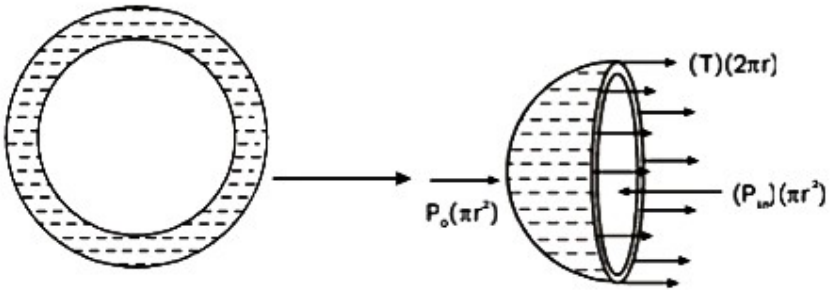


Pressure inside the liquid bubble $= P_0 + \dfrac{4T}{r}$

So, excess pressure inside the liquid bubble $= \dfrac{4T}{r}$

Alternative method:

Draw free body diagram of half part of bubble. The force on this hemisphere are:



(i) Pushing force on the left half liquid due to right half liquid will be $(P_{in})(\pi r^2)$

(ii) Pushing force due to atmospheric pressure will be (P_0) × (facing area) $= P_0(\pi r^2)$

(iii) Surface tension force on both inner and outer surface will be $(T)(2\pi r) \times 2$

Applying force balance:

$$P_{in}\left(\pi r^2\right) = \left[(T)(2\pi r) \times 2 + P_0(\pi r^2)\right] \Rightarrow P_{in} = P_0 + \frac{4T}{r},$$

So, excess pressure inside a liquid bubble $= \dfrac{4T}{r}$

Q. **Radius of bubble B is slightly greater than radius bubble A. If we open the stopper, in which direction air will flow and which of following will remain constant?**

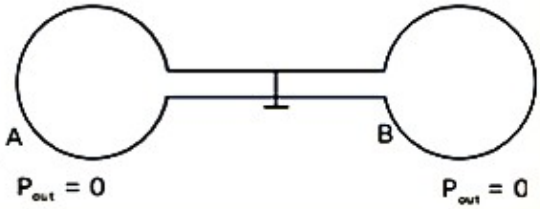
(i) **Total volume of two bubbles.**

(ii) **Total surface area of two bubble.**

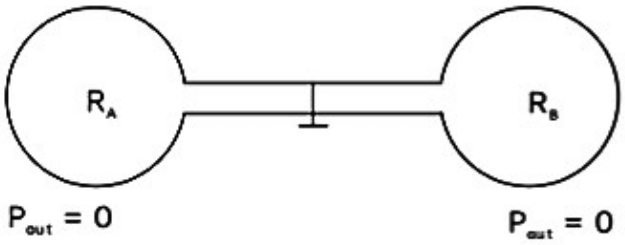
(iii) **Total no. of moles in two bubbles.**

(iv) **Total surface energy in two bubbles.**

Assume, air flows slowly.



Sol. Since, radius of bubble B is slightly greater than radius of bubble A. Therefore, pressure inside bubble B ($4T/R_B$) will be less then pressure inside bubble A ($4T/R_A$) and air will flow from A to B. As a result of this radius of bubble B keep on increasing and radius of bubble of A keep on decreasing till pressure becomes equal.



Since, number of molecules during this process are constant

So, $n_A + n_B = n$

$$\Rightarrow \quad \frac{P_A V_A}{RT} + \frac{P_B V_B}{RT} = \frac{PV}{RT}$$ (Where R is universal gas constant)

$$\Rightarrow \quad \frac{P_A \cdot \frac{4}{3}\pi R_A^3}{RT} + \frac{P_B \cdot \frac{4}{3}\pi R_B^3}{RT} = \frac{P \cdot \frac{4}{3}\pi R'^3}{RT}$$

$$\Rightarrow \quad P_A R_A^3 + P_B R_B^3 = PR'^3$$

$$\frac{4T}{R_A} \times R_A^3 + \frac{4T}{R_B} \times R_B^3 = \frac{4T}{R'} \times R'^3$$

$$R_A^2 + R_B^2 = R'^2$$

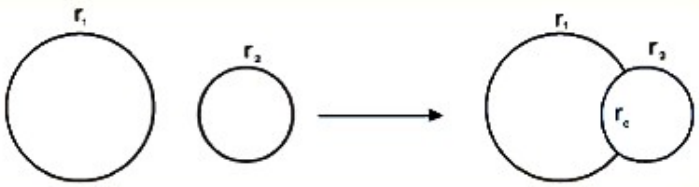
$$4\pi R_A^2 + 4\pi R_B^2 = 4\pi R'^2$$

$\Rightarrow$ Surface area remains constant during the process.

$$4\pi T R_A^2 + 4\pi T R_B^2 = 4\pi T R'^2$$

Hence, Surface energy remains constant during this process.

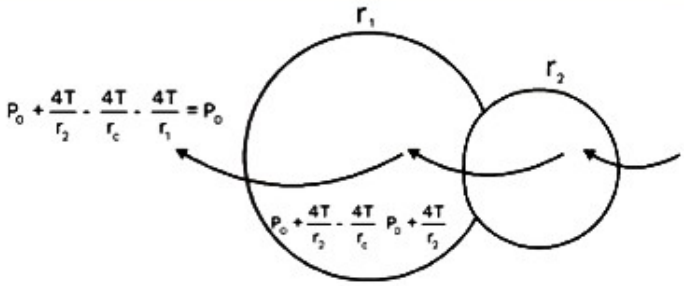
Q. **Two soap bubbles of radius r_1 and r_2 combine. Find radius of curvature of the common surface separating them.**



Sol. $$P_0 + \frac{4T}{r_2} - \frac{4T}{r_c} - \frac{4T}{r_1} = P_0$$

$$\frac{1}{r_c} = \frac{1}{r_2} - \frac{1}{r_1}$$

$$r_c = \frac{r_1 r_2}{r_1 - r_2}$$



Q. **A soap bubble of radius r and surface tension constant T is given a charge, so that its surface charge density is σ. Due to charge, the radius of the soap bubble becomes double then find 'σ'. (atmospheric pressure = P_0)**

Sol. Initial pressure inside the bubble $P_i = P_0 + \frac{4T}{r_i}$.

Now a uniform surface charge in given to the bubble. The surface tension is a pulling force, which increases pressure inside the bubble (by $\frac{4T}{r}$), but the charges given to the surface will repel each other. So due to the charge given, pressure inside the bubble will decrease (by $\frac{\sigma^2}{2\varepsilon_0}$).

So, final pressure inside the bubble

$$P_f = P_0 + \frac{4T}{r_f} - \frac{\sigma^2}{2\varepsilon_0}$$

As the temperature inside the bubble is constant so,

$$P_i V_i = P_f V_f$$

$$\left(P_0 + \frac{2T}{r}\right)\left(\frac{4}{3}\pi r^3\right) = \left(P_0 + \frac{4T}{r_f} - \frac{\sigma^2}{2\varepsilon_0}\right)\left(\frac{4}{3}\pi r_f^3\right)$$

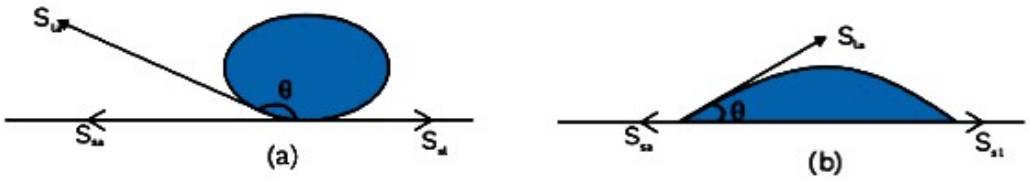
Here put $r_f = 2r$

So, we get $\sigma = \sqrt{\left(7P_0 - \frac{12T}{r}\right)2\varepsilon_0}$

CONTACT ANGLE

The surface of liquid near the plane of contact, with another medium is in generally curved. *The angle between tangent to the liquid surface at the point of contact and solid surface inside the liquid is termed as angle of contact.* It is denoted by θ.

1. It is different at interfaces of different pairs of liquids and solids.
2. The value of θ determines whether a liquid will spread on the surface of a solid or it will form droplets on it. For example, water forms droplets on lotus leaf as shown in Fig. (a) while spreads over a clean plastic plate as shown in Fig. (b).



We consider the three interfacial tensions at all the three interfaces, liquid-air, solid-air and solid-liquid denoted by S_{la}, S_{sa} and S_{sl} respectively as given in Fig. (a) and (b). At the line of contact, the surface forces between the three media must be in equilibrium.

$$S_{la}\cos\theta + S_{sl} = S_{sa}$$

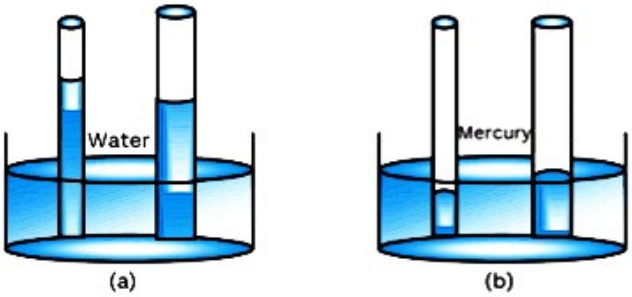
Case-1: The angle of contact is an obtuse angle- if $S_{sl} > S_{la}$ as in the case of water-leaf interface liquids are attracted strongly to themselves and weakly to those of solid, it costs a lot of energy to create a liquid-solid surface, and liquid then does not wet the solid. This is what happens with water on a waxy or oily surface, and with mercury on any surface.

Case-2: The angle of contact is an acute angle- if $S_{sl} < S_{la}$ as in case of water-plastic interface. In this case the molecules of the liquid are strongly attracted to those of the solid, this will reduce S_{sl} and therefore, $\cos\theta$ may increase or θ may decrease. In this case θ is an acute angle. This is what happens for water on glass or on plastic and for kerosene oil on virtually anything (it just spreads).

Soaps, detergents and dying substances are wetting agents. When they are added the angle of contact becomes small so that these may penetrate well and become effective. Water proofing agents on the other hand are added to create a large angle of contact between the water and fibres.

Capillary action

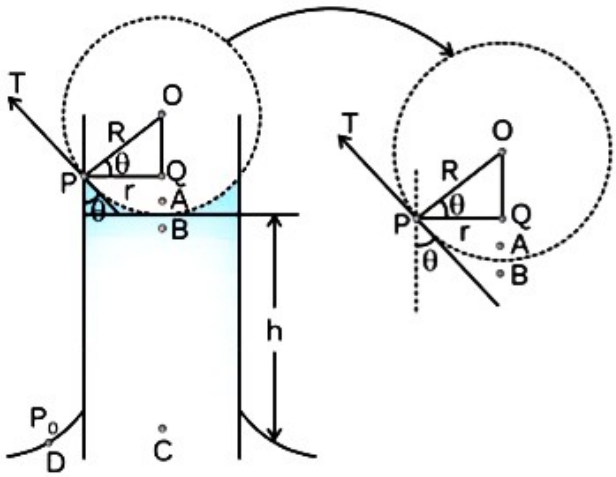
If a narrow glass tube open at both ends is pushed in water, water rises in the tube to a height above it's surface.



Capillary rise or fall depends on contact angle between liquid and tube

Rise of Liquid in a Capillary Tube

Four points A, B, C and D as shown in the figure. A and B are the points just above and just below the meniscus, respectively.



Now, we have

$$P_B = P_0 - \frac{2T}{R} \quad \text{...(i)}$$

$$P_B = P_0 - \rho gh \quad \text{...(ii)}$$

Using Equation (i) and (ii), we have

$$P_0 - \rho gh = P_0 - \frac{2T}{R}$$

This gives $h = \frac{2T}{\rho gR}$

From ΔOPQ, $R = \frac{r}{\cos\theta}$

Then $\boxed{h = \frac{2T\cos\theta}{\rho g r}}$ .....**Ascent formula**

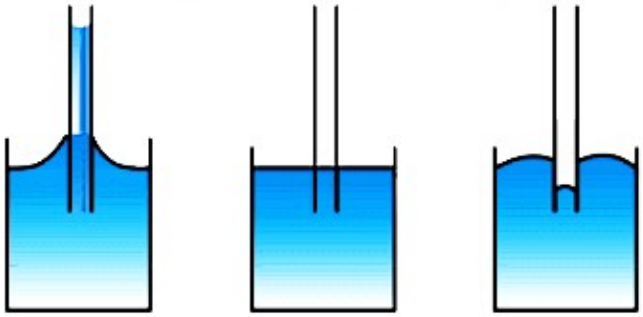
Note:

From this formula, we can say that

If $\theta < 90°$ then $h > 0$, so water in capillary will rise up (Ex. water in glass tube).

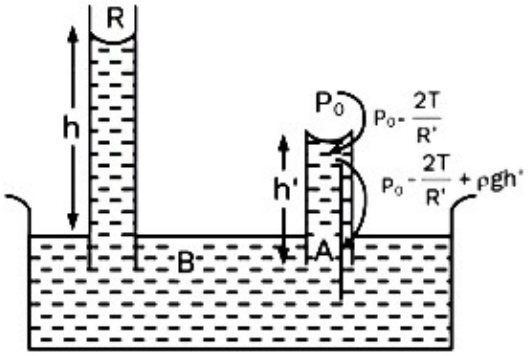
If $\theta = 0$ then $h = 0$, so water in the capillary will not rise. (Ex. water in silver tube).

If $\theta > 90°$ then $h < 0$,so liquid in capillary will go down. (Ex. mercury in glass tube.)



Rise of liquid in a capillary tube of insufficient length:

If the length of the tube h′ is less than h, it is found that the liquid does not overflow. In a tube of insufficient length, the liquid rises up to the top of the tube and increases the radius of curvature of its meniscus to a value R′,



Pressure inside the meniscus $P=P_0-2T/R'$

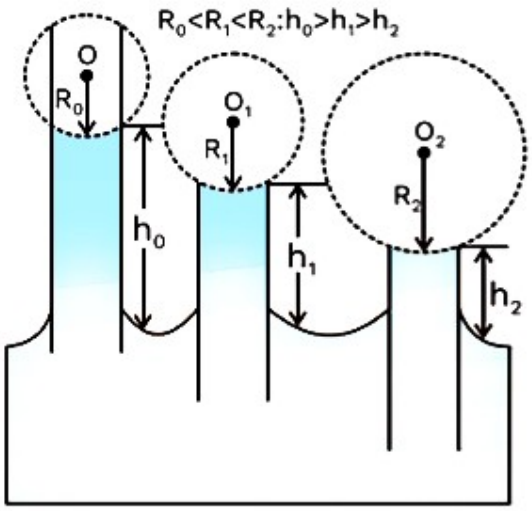
Pressure inside the tube at water surface ($P_0-2T/R' + \rho g h'$) will be equal to P_0

$\Rightarrow P_0 - 2T/R' + \rho g h' = P_0$

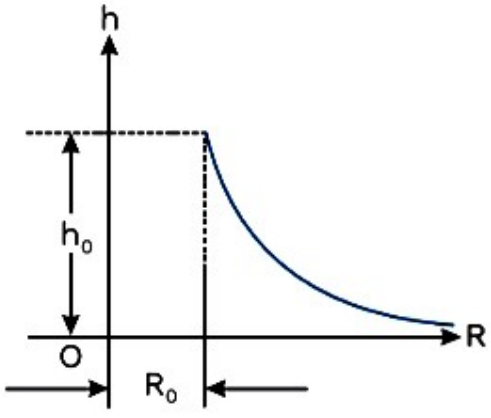
$\Rightarrow hR' = \frac{2T}{\rho g} = a(\text{constant})$

$\Rightarrow \quad R'h' = Rh$

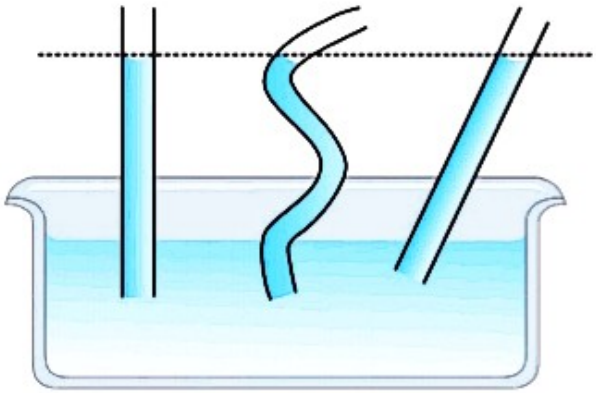
That is smaller the length (h') of the tube, greater will be the radius of curvature (R') of the meniscus but liquid will never overflow.



$$\boxed{\frac{h_1}{h_2} = \frac{R_2}{R_1} = \frac{\cos\theta_1}{\cos\theta_2}}$$



Note: The height of water in the capillary $\left(h = \frac{2T}{\rho g r}\cos\theta\right)$ doesn't depend on shape of the capillary. So, water will raise to same height in all the tubes.
(However, the length of water column in the tubes can be different)



Practical Applications of Capillary

1. The oil in a lamp rises in the wick by capillary action.
2. Water rise upto the top of the leaves of the tree by capillary action.
3. If one end of the towel dips into a bucket of water and the other end hangs over the bucket the towel soon becomes wet throughout due to capillary action.
4. Ink is absorbed by the blotter due to capillary action.
5. Sandy soil is more dry than clay. It is because the capillaries between sand particles are not so fine as to draw the water up by capillaries.
6. The moisture rises in the capillaries of soil to the surface, where it evaporates. To preserve the moisture in the soil, capillaries must be broken up. This is done by ploughing and levelling the fields.
7. Bricks are porous and behave like capillaries.

Q. **A glass rod of diameter d_1 = 1.5 mm is inserted symmetrically into a glass capillary with inside diameter d_2 = 2.0 mm. Then the whole arrangement is vertically oriented and brought in contact with the surface of water. To what height will the liquid rise in the capillary? Surface tension of water = 73 × 10^{-3} N/m.**

Sol. If R is radius of meniscus, then $\frac{2T}{R} = h\rho g$

Here $R = \frac{r_2 - r_1}{\cos\theta}$

θ being angle of contact, r_1 = radius of glass rod, r_2 = radius of capillary.

$$\frac{2T\cos\theta}{r_2 - r_1} = h\rho g$$

$$h = \frac{2T\cos\theta}{(r_1 - r_1)\rho g}$$

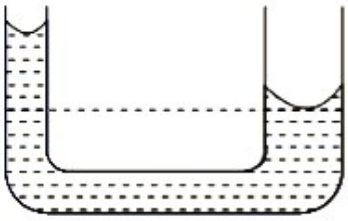
Here $r_1 = \frac{d_1}{2}, r_2 = \frac{d_2}{2}$

$$\therefore h = \frac{4T\cos\theta}{(d_2 - d_1)\rho g}$$

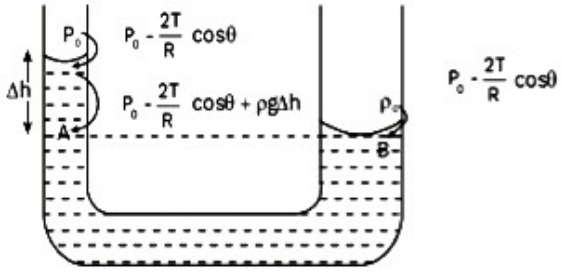
Substituting given values and $\theta \cong 0°$ for water-glass interface, we have

$$h = \frac{4 \times 73 \times 10^{-3} \cos 0°}{(2.0 - 1.5) \times 10^{-3} \times 10^3 \times 9.8} = 60 \times 10^{-3} m = 6cm$$

Q. **In the U-tube, radius of one arm is R and the other arm is 2R. Find the difference in water level if contact angle is $\theta = 60°$ and surface tension of water is T.**



Sol. Balancing pressure at points A and B situated in same horizontal level.



$$P_0 - \frac{2T}{R}\cos\theta + \rho g\Delta h = P_0 - \frac{2T}{2R}\cos\theta$$

here $\theta = 60^0$, solving we get

$$\Delta h = \frac{T}{2\rho g R}$$

Q. **There is a small hole of diameter 0.1 mm at the bottom of a large container. To what minimum height we can fill water in it, so that water doesn't come out of hole. (T_{water} = 75 × 10^{-3} N/m) ρ_{water} = 10^3 kg/m^3, g = 10m/sec^2.**

Sol. The lower surface of water, which will try to come out will be spherical. Pressure just outside the spherical surface is:

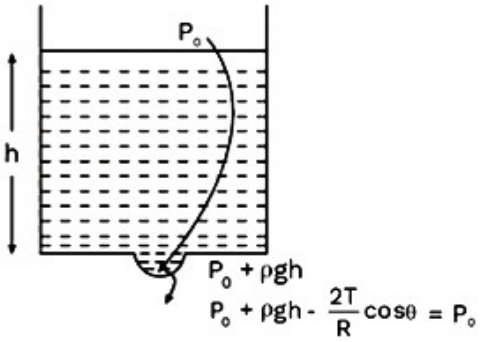
$$P_0 + \rho gh - \frac{2T}{R}\cos\theta = P_0$$

$$h = \frac{2T}{\rho gR}\cos\theta$$

$$(h)_{max} = \frac{2T}{\rho gR}(\cos\theta)_{max} \text{ and } (\cos\theta)_{max} = 1$$

So, $$(h)_{max} = \frac{2T}{\rho gR} = \frac{2 \times 75 \times 10^{-3}}{10^3 \times 10 \times 0.1 \times 10^{-3}}$$

$$(h)_{max} = 15\,cm$$



SOME OTHER APPLICATIONS OF SURFACE TENSION

(i) The wetting property is made use of in detergents and waterproofing. When the detergent materials are added to liquids, the angle of contact decreases and hence the wettability increases. On the other hand, when water proofing material is added to a fabric, it increases the angle of contact, making the fabric water-repellent.

(ii) The antiseptics have very low value of surface tension. The low value of surface tension prevents the formation of drops that may otherwise block the entrance to skin or a wound. Due to low surface tension the antiseptics spreads properly over the wound. The lubricating oils and paints also have low surface tension. So, they can spread properly.

(iii) Surface tension of all lubricating oils and paints is kept low so that they spread over area.

(iv) Oil spreads over the surface of water because the surface tension of oil is less than the surface tension of cold water.